

Chengpo Yan

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EDUCATION

University of Wisconsin-Madison

Ph.D. in Computer Science

Madison, WI

Sep. 2023 – May 2028

University of Wisconsin-Madison

B.S. in Computer Science & B.S. in Data Science

Madison, WI

Sep. 2019 – Dec. 2022

RESEARCH INTERESTS

My research interests are in **Systems** and **Machine Learning**. My current focus is on optimizing the inference system of LLMs to improve throughput on cloud platforms and edge devices.

PUBLICATIONS

- [1] BagPipe: Accelerating Deep Recommendation Model Training. Saurabh Agarwal, **Chengpo Yan**, Ziyi Zhang, Shivaram Venkataraman. [SOSP 2023]
- [2] Deep-Learning-Based Segmentation of Keyhole in In-Situ X-ray Imaging of Laser Powder Bed Fusion. William Dong, Jason Lian, **Chengpo Yan**, Yiran Zhong, Sumanth Karnati, Qilin Guo, Lianyi Chen, Dane Morgan. [Materials 2024]
- [3] Privacy-Aware Sharing of Raw Spatial Sensor Data for Cooperative Perception. Bangya Liu, **Chengpo Yan**, Chenghao Jiang, Suman Banerjee, Akarsh Prabhakara. [HotMobile 2026]

RESEARCH EXPERIENCE

Research Assistant with Prof. Suman Banerjee

Sep. 2024 – Present

Deployment of Large Language Models on Tiny Devices

- Designed a token-driven, cross-accelerator inference system for vision-language models running on mobile SoCs.
- Optimized GPU-accelerated GEMM kernel on unified memory architecture, reducing the memory usage by 11.2%.
- Reduced end-to-end inference latency by 36.2% with on-chip GPU and NPU.

Undergraduate Research Assistant with Prof. Shivaram Venkataraman

May 2022 – Dec. 2022

BagPipe: Accelerating Deep Recommendation Model Training

- Systematically arranged and cached the embeddings required for a Deep Learning Recommendation Model over distributed GPU clusters to optimize the time for fetching and updating embeddings given a parameter server training cluster.
- Reduced overall training time by 10% by overlapping the forward pass with NCCL-based cache synchronizations.
- Developed a policy for writing the embeddings evicted from cache on each worker to a parameter server under distributed GPU clusters, reducing the IO by 15%.
- Applied BagPipe to other Deep Recommendation Models, including W&D, D&C, DEEPPFM, FGCNN, and CASER.
- Achieved up to 5x efficiency gain compared to the original DLRM proposed by Meta.

Undergraduate Research Assistant with Prof. Suman Banerjee

Jul. 2022 – Dec. 2023

Wi-Fi & Bluetooth Based Indoor Localization

- Designed a localization system that can use either RSSI or CSI, common indicators from Wi-Fi and Bluetooth devices.
- Tested different localization algorithms, including kNN, Stg, CSE, and Gaussian Kernel, achieving a median accuracy of two meters with the Gaussian Kernel approach.
- Used Apple's Airport to collect and generate RSSI dataset for the UW-Madison CS Department, which includes over a hundred access points.

Undergraduate Researcher with Prof. Dane Morgan @ Informatics Skunkworks

Mar. 2022 – Dec. 2022

Identifying Defect Formation Mechanisms from X-ray Imaging Data with Machine Learning

- Designed a continuous workflow of the entire identification and measurement stream, including image labeling and image measuring.
- Deployed BASNet, a boundary-aware segmentation network to label raw images for measurements, which shows promising accuracy of 91.24% for keyhole area and 92.81% for boundary shape.
- Implemented a parallel-based measuring tool for retrieving specific quantities given labeled data.

PROFESSIONAL EXPERIENCE

University of Wisconsin-Madison

Teaching Assistant

Madison, WI

Sep. 2023 – Present

- Spring 2024 to Spring 2026: CS354 Machine Organization and Programming.
- Fall 2023: CS540 Introduction to Artificial Intelligence.

Skyworth

Software Engineer Intern

Shenzhen, China

May 2021 – Aug. 2021

- Optimized Swaiot application, an EEUI-based cross-platform application for controlling indoor devices through Skyworth AIOT system, achieving a 9% improvement in loading time and 11% in response time.
- Developed a theme-switching functionality for Swaiot application, enabling the user to switch between light and dark mode.
- Adapted Swaiot application for various Skyworth devices, including Skyworth remote control, Skyworth tablet, and Skyworth smart table.

University of Wisconsin-Madison

Web Development Club

Madison, WI

Sep. 2019 – May 2022

- Contributed to the design of the UW-Madison CS Department's webpage, making it more user-friendly.
- Tutored web development frameworks and techniques.

PROJECTS

Matrix Calculator

Mar. 2020 – May 2020

An Application for Matrix Calculation

- Implemented a JavaFX-based desktop application to provide accessible matrix calculation services for students.
- Planned the general workflow and designed the appearance of the application.

Mini-Twitter

May 2020 – Jun. 2020

A Mini Web Service that Emulates Twitter

- Implemented the web service with Ruby, Rails, MongoDB, Webpacker for online posting service.
- Developed Ajax-based frontend using ReactJS, Redux, and Saga to perform real-time notifications.

UniMatch

Jun. 2020 – Aug. 2020

A Web Service that Offers College Information and Provides Suggestions

- Implemented the web service with ReactJS, Webpacker, Flask, TensorFlow for matching up to five universities.
- Developed Python-based backend with Flask and TensorFlow to find the most suitable university.
- Continuous deployment and integration with TravisCI.

RELEVANT COURSES

Computer Science: Algorithms, Artificial Intelligence, Computer Architecture, Data Analysis, Networks, Operating Systems, Human-Computer Interaction, Virtual Reality

Mathematics: Calculus, Discrete Math, Math in Data Science, Numerical Linear Algebra, Probability

Statistics: R in Data Science, Random Variables

TECHNICAL SKILLS

Programming: Java, Python, C, C#, C++, JavaScript, HTML/CSS, R

Frameworks: PyTorch, CUDA, NCCL, SGLang, vLLM, DeepSpeed, Docker

AWARDS

2018 Meritorious Award in Mathematical Contest in Modeling
Dean's List from 2019 to 2022 (6 semesters)